SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Scenario-Based Questions

SET 3 :

1. A manufacturing company plans to migrate its legacy enterprise applications and databases from an on-premises data center to the cloud. The organization wants to minimize downtime while ensuring application performance and security.

- Design a cloud migration strategy suitable for the organization.
- Explain how different migration approaches (Rehost, Refactor, Replatform, Repurchase, Retire, Retain) can be applied.
- Discuss methods to validate application performance after migration.
- Recommend strategies to reduce migration risks and downtime.

2. A smart city project deploys thousands of IoT sensors to monitor traffic, air quality, energy consumption, and public safety. The collected data must be processed and analyzed in real time for decision-making.

- Design a cloud-based architecture for IoT data collection and analytics.
- Explain how message brokers, stream processing, and cloud storage are integrated.
- Discuss methods for handling high-volume, high-velocity sensor data.
- Recommend techniques to ensure scalability, reliability, and security.

3. A global e-commerce company wants executives to monitor sales performance, customer behavior, and regional trends through interactive dashboards. The solution must support real-time analytics and historical reporting.

- Design a cloud-based business intelligence architecture.
- Explain how data visualization tools integrate with cloud data warehouses.
- Discuss techniques for optimizing dashboard performance with large datasets.
- Recommend security measures to protect business reports and sensitive data.

4. An online gaming company experiences fluctuating workloads throughout the day. During peak hours, users report high latency and reduced application performance.

- Design a cloud monitoring and performance optimization solution.
- Explain how cloud monitoring tools, metrics, logs, and tracing help identify performance bottlenecks.
- Discuss strategies for resource optimization and capacity planning.
- Recommend methods to improve application availability while controlling operational costs.

5. A financial institution stores petabytes of customer transaction data for regulatory compliance and business analytics. The organization must retain historical records while managing storage costs efficiently.

- Design a cloud-based data lifecycle management strategy.
- Explain how data classification, retention policies, and archival storage should be implemented.
- Discuss the importance of governance, auditing, and regulatory compliance in big data environments.
- Recommend techniques for balancing data accessibility, security, and cost optimization.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1) Cloud Migration of Legacy Enterprise Applications.

(i) Cloud Migration strategy:

The manufacturing company can use a phased cloud migration strategy to move its legacy applications and databases from the on-premises data center to the cloud with minimum downtime.

Assessment:

Identify applications, databases, dependencies, and performance requirements.

Planning:

Select suitable cloud services and migration tools.

Pilot Migration:

Migrate a small application to test the process.

Data Migration:

Transfer databases securely with continous replication.

Application Migration:

Move applications in planned phases.

Testing:

Check application functionality, performance, and security.

Deployment:

Switch users to the cloud after successful validation.

Different Migration Approaches:

(i) Rehost
(ii) Refactor
(iii) Replatform
(iv) Repurchase
(v) Retire
(vi) Retain

Application Performance Validation:

- Conduct load and stress testing.
- Compare response times before and after migration.
- Monitor CPU, memory, network, and database perfo
  — traci
- Use cloud monitoring tools to identify bottlenecks.

Reducing Migration Risks and Downtime:

- Maintain backups before migration.
- Use incremental data replication to keep databases synchronized.
- Migrate application in phases.
- Apply encryption, Access controls & Security
  — testing.

Conclusion:

A phased Migration with Continuous replication, proper testing and rollback planning reduces downtime, migration risks and performance issues.

2)

(i) Cloud-Based IoT Architecture:

A Smart city can use cloud Computing to collect Process, store and Analyze data from thousands of IoT Sensors.

- IoT Sensors
- IoT Gateway
- Message Broker
- Stream processing
- Cloud Storage
- Analytics Dashboard.

(ii) Integration of Message Brokers, Stream Processing and Storage.

- Message brokers such as MQTT brokers or Apache Kafka receive Sensor Messages.
- Stream processing tools analyze incoming data continuously
- Dashboards display alerts and important immediately.

(iii) Handling High-Volume Sensor Data

- Use distributed storage and parallel processing.
- Filter unnecessary data before storage.
- Compress data to reduce storage requirments.
- Use Edge Computing for Preliminary processing near Sensors.

(iv) Scalability, Reliability and Security:

- Use auto-scaling to handle increasing sensor traffic.
- Deploy services across multiple availability zones.
- Encrypt data during transmission and storage.
- Apply Authentication and Access Controls.
- Monitor system health and configure automatic Alerts.

Conclusion:

Cloud-based IoT analytics enables smart cities to make faster decisions, improve public services, and manage resources efficiently.

3)

(i) Cloud-Based Business Intelligence Architecture

A cloud-based Business Intelligence (BI) system helps executives analyze sales, customer behavior and regional trends.

* Data Sources.
* Data Ingestion
  * ETL/ELT pipeline
  * Cloud Data Warehouse
  * BI Tools:
  * Decision-Making

2) Integration of Visualization Tools and Cloud Data Warehouses.

- Connect BI tools to cloud data warehouses.
- Retrieve processed data using SQL Queries.
- Use filters and drill-down features for detailed analysis.

3) Optimizing Dashboard Performance

- Use Indexes and Partitioning where supported.
- Optimize SQL queries and Avoid unnecessary Columns.
- Apply caching for frequently accessed reports.

4) Security Measures

* Implement role-based Access Control.
* Encrypt sensitive data at rest and in transit.
* Apply data masking for confidential customer information.
* follow applicable data protection regulations.

Conclusion:

A cloud-based BI architecture provides faster reporting, interactive dashboards, and secure business.

4) (i) Cloud Monitoring and Optimization Solution

The Online gaming Company can use cloud monitoring and auto-scaling to reduce latency and maintain applications performance during peak hours.

* Monitoring Tools
* Metrics
* Logs
* Distributed Tracing
* Load Balancer
* Auto-Scaling

(ii) Role of Metrics, Logs and Tracing

- Metrics identify high resource usage and slow response times.
- Logs help detect errors, crashes and failed requests.
- Tracing follows request across multiple servics
- Monitoring dashboards helps identify and resolve bottlenecks quicky.

(iii) Resource Optimization and Capacity Planning.

- Analyze historical traffic and peak-hour workloads.
- Use auto-scaling to match resources with demand.
- Use reserved capacity for predictable workloads and On-demand resources for sudden demand.

(iv) Improving Availability and Controlling Costs

- Deploy game servers across multiple availability zones.
- Use load balancing and health checks.
- Maintain backups and disaster recovery plans.
- Use reserved capacity for predictable workloads and on-demand resources for sudden demand.
- Perform regular load and stress testing.

Conclusion:

Cloud monitoring, auto-scaling, and resource optimization improve gaming performance and control operational costs.

5)

(i) Cloud-Based Data Lifecycle Management Strategy.

The financial Institution can implement cloud data lifecycle management to retain petabytes of transaction data securely while reducing storage costs.

* Data Collection
* Data Classification:
* Active Storage
* Archival Storage
* Retention.

(ii) Data Classification, Retention and Archival

* Classify data as public, internal, confidential, or highly sensitive.

* Define retention periods according to legal and Business requirements.

* Keep frequently accessed data in faster storage.

(iii) Governance, Auditing & Regulatory Compliance

* Establish data governance policies and Ownership

* Maintain Accurate metadata and data Catalogues

* Conduct regular Compliance audits

* Follow applicable financial regulations and data protection requirements.

(iv) Balancing Accessibility, Security and Cost

* Use Storage tiers according to data access frequency.

* Compress and deduplicate data where appropriate.

* Use strong authentication and least-privilege access.

* Test backup restoration and Archival Retrieval regularly.

Conclusion:

Cloud data lifecycle management ensures secure data retention, regulatory Compliance, Easy retrieval of required records, and efficient Storage-cost management.